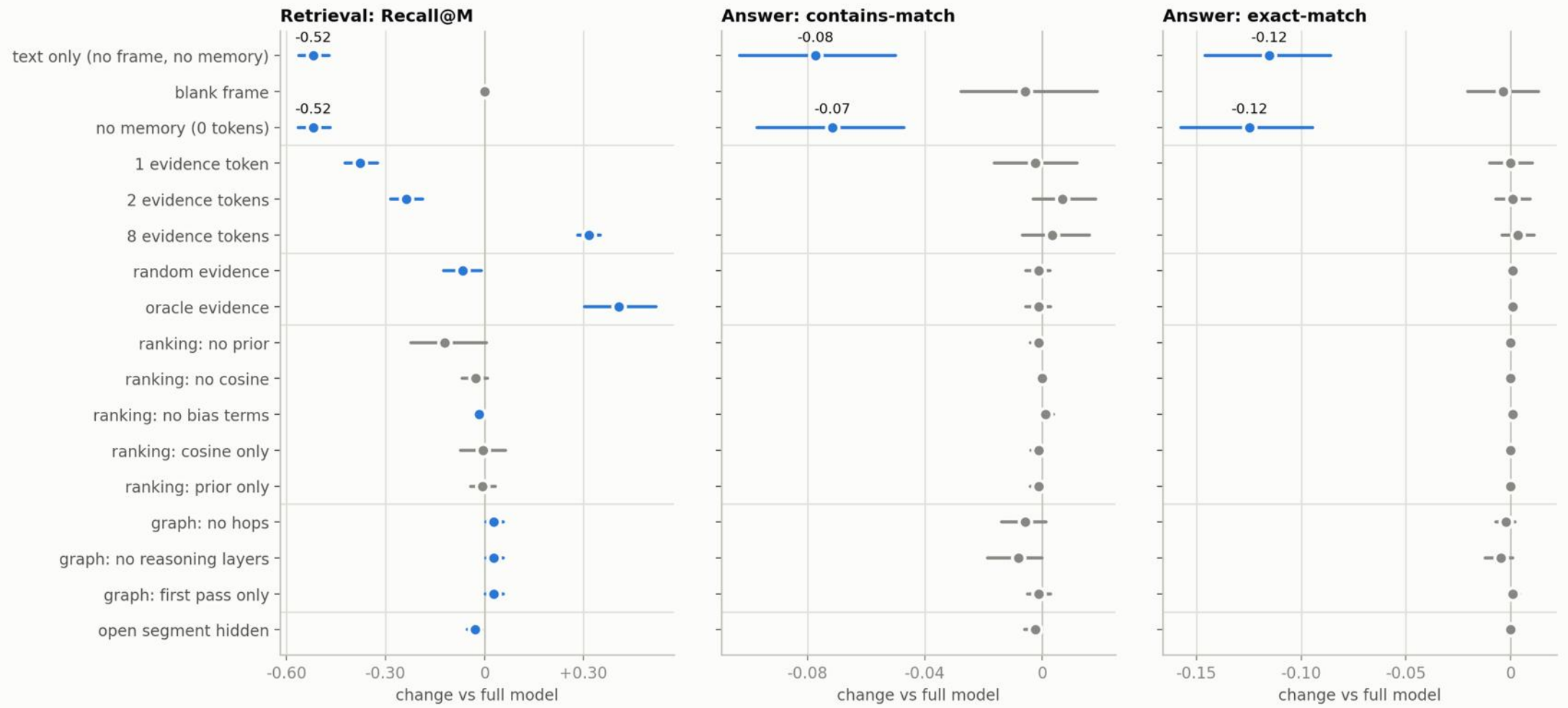
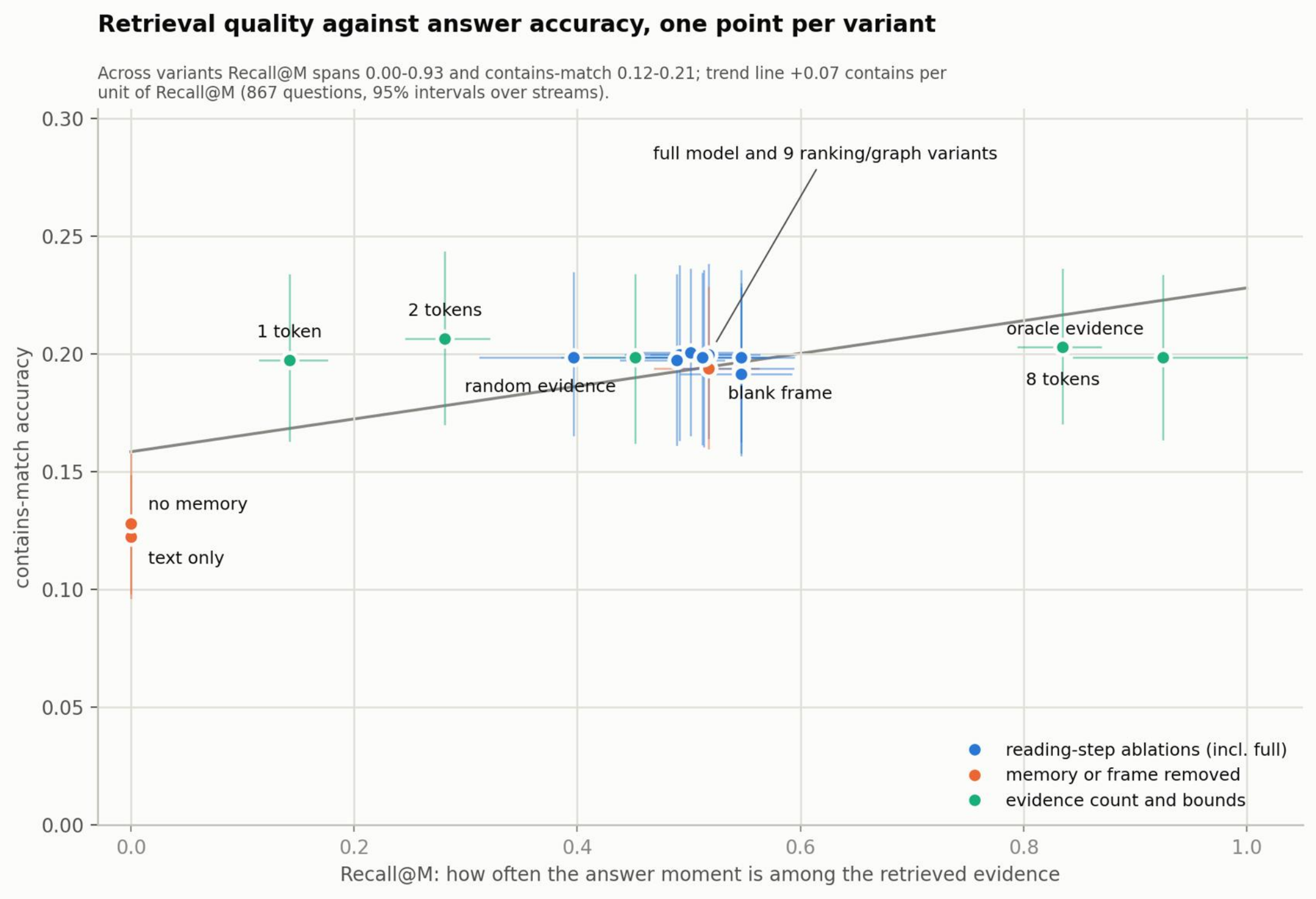
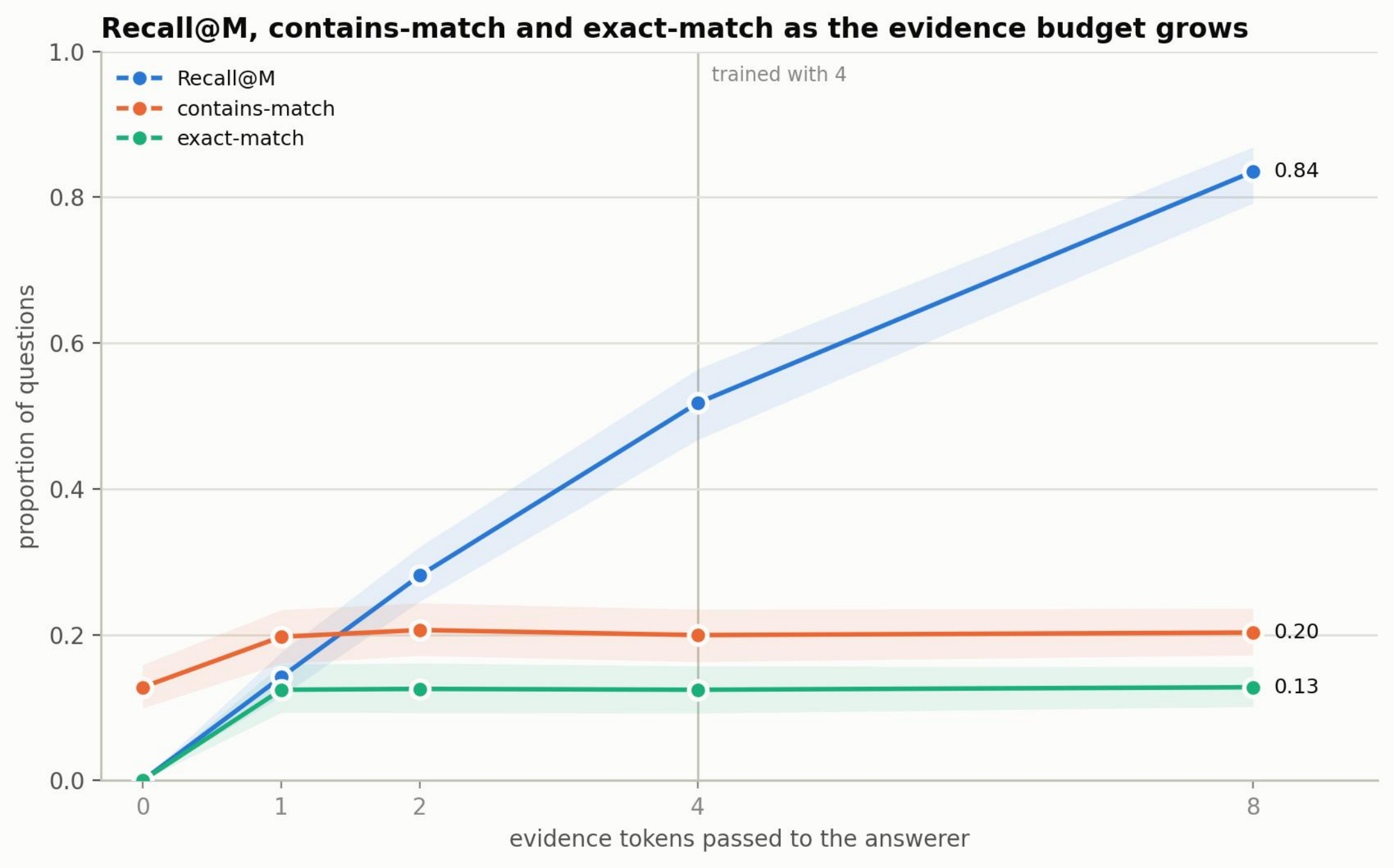
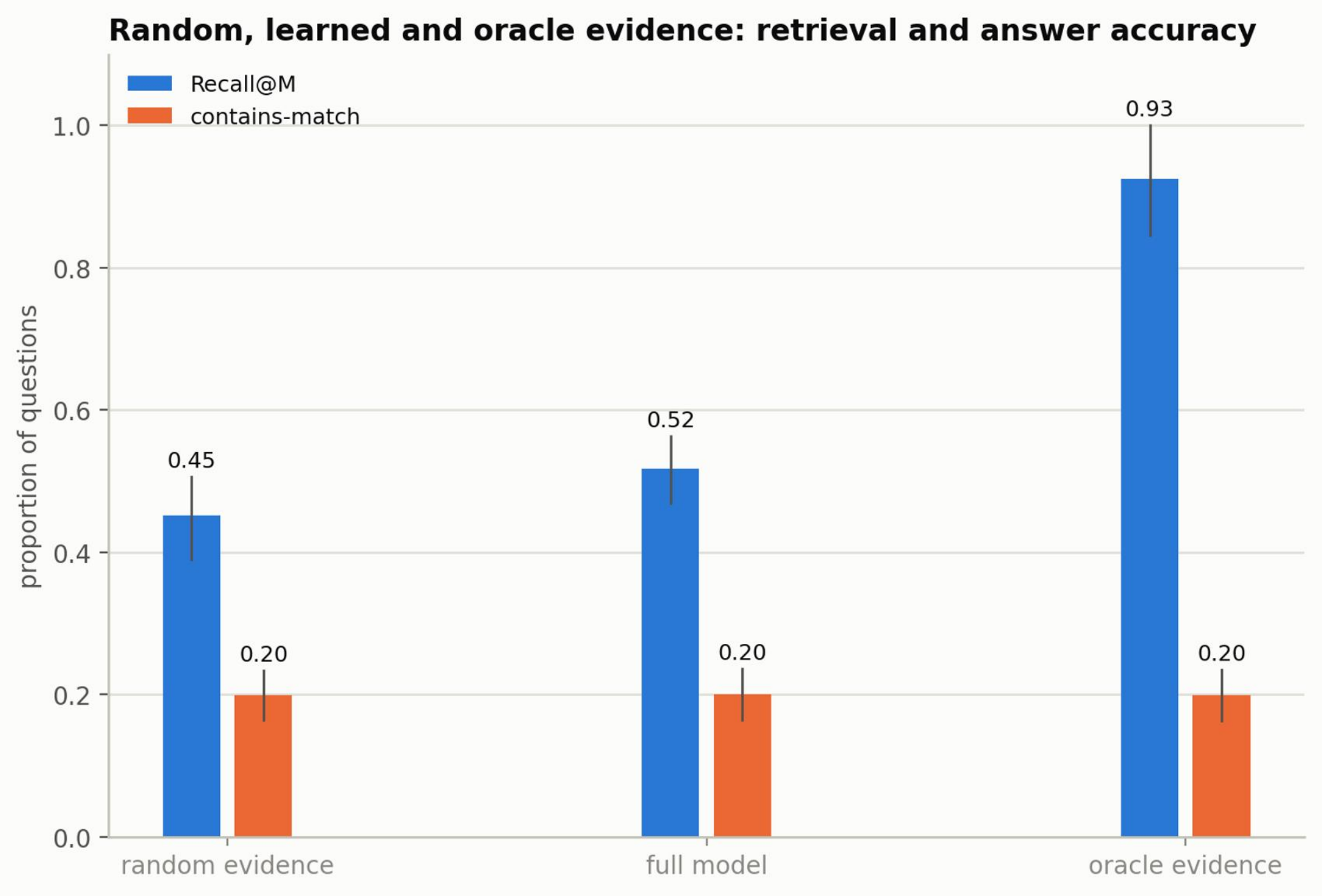
What changes when one part of the reader is altered (variant minus full model)

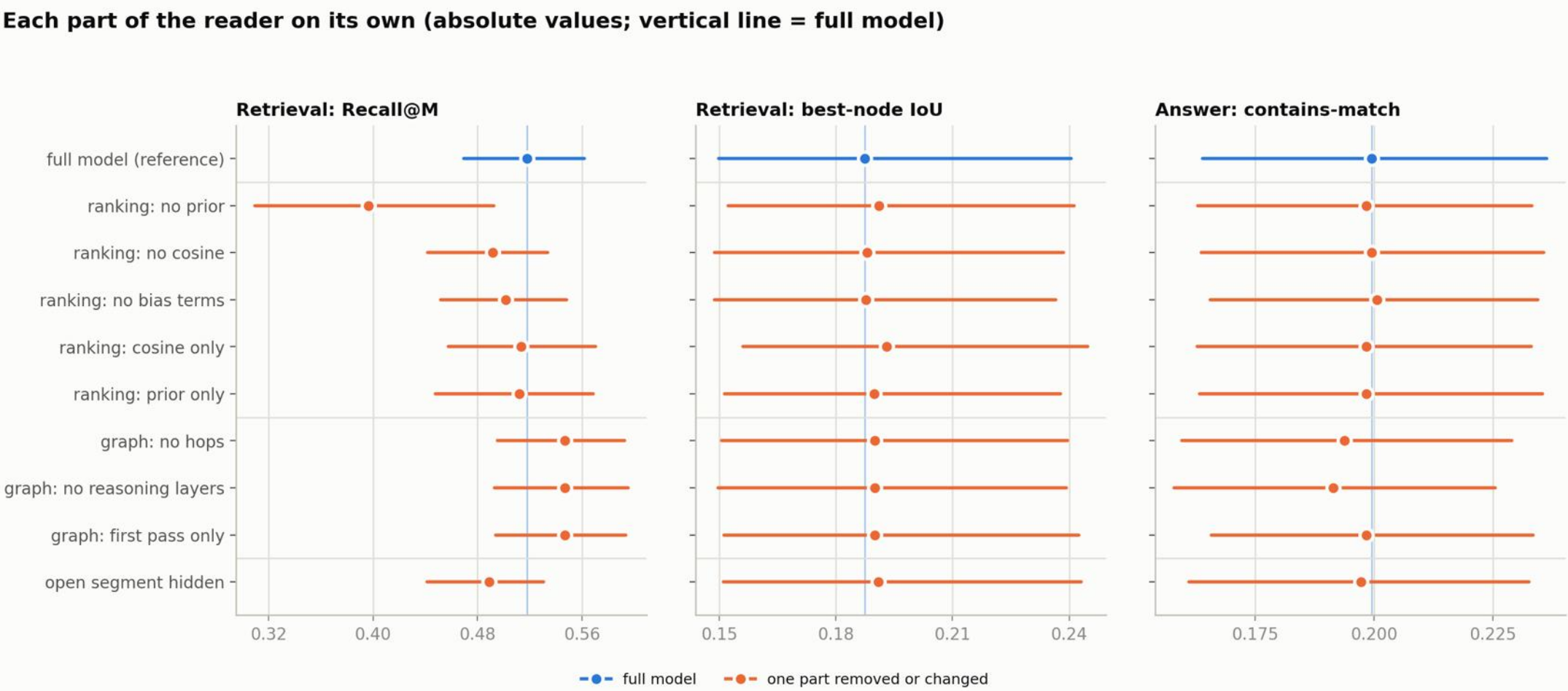
Largest change: Recall@M -0.52, contains -0.08, exact -0.12. 867 questions, 50 validation streams; bars are 95% bootstrap intervals over streams.




Retrieval quality against answer accuracy, one point per variant
Across variants Recall@M spans 0.00-0.93 and contains-match 0.12-0.21; trend line +0.07 contains per unit of Recall@M (867 questions, 95% intervals over streams).
full model and 9 ranking/graph variants
1 token
2 tokens
random evidence
blank frame
oracle evidence
8 tokens
no memory
text only
reading-step ablations (incl. full)
memory or frame removed
evidence count and bounds
contains-match accuracy
Recall@M: how often the answer moment is among the retrieved evidence
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.0
0.2
0.4
0.6
0.8
1.0

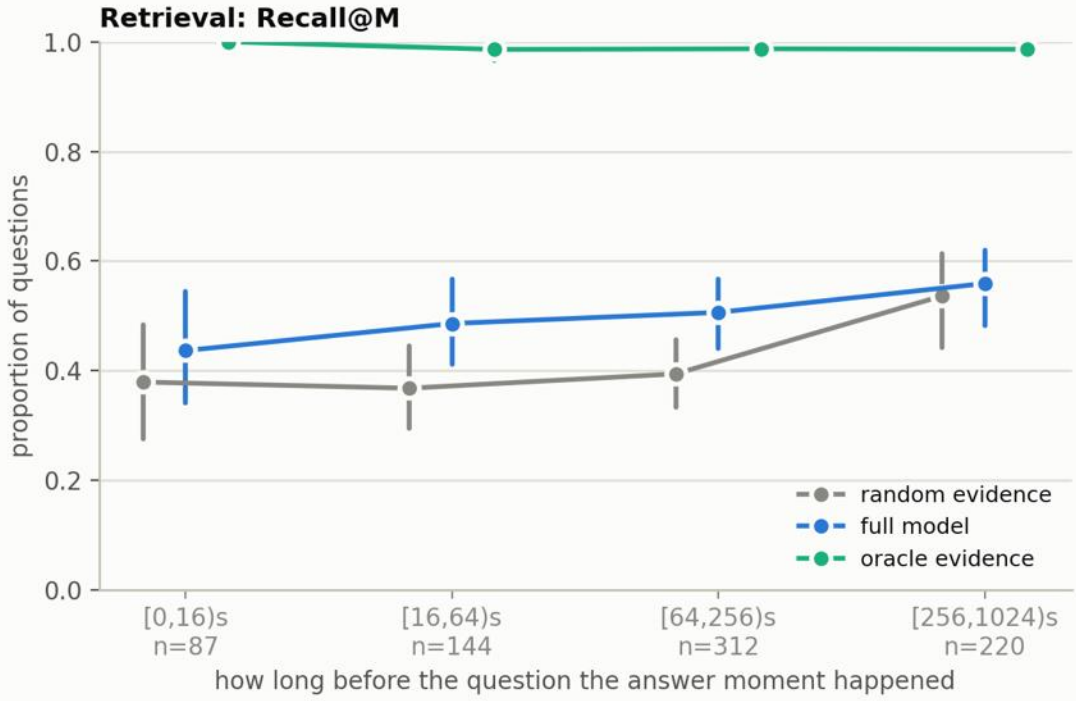

Recall@M, contains-match and exact-match as the evidence budget grows
Recall@M
contains-match
exact-match
trained with 4
0.84
0.20
0.13
1.0
0.8
0.6
0.4
0.2
0.0
proportion of questions
0
1
2
4
8
evidence tokens passed to the answerer

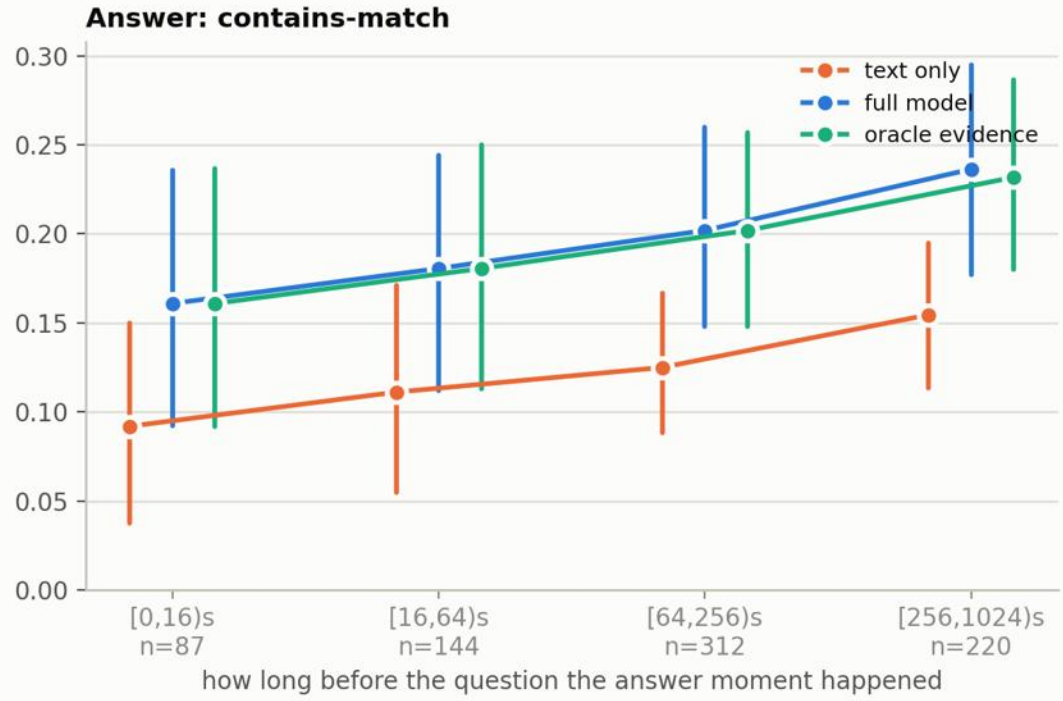

Random, learned and oracle evidence: retrieval and answer accuracy
Recall@M
contains-match
proportion of questions
1.0
0.8
0.6
0.4
0.2
0.0
0.45
0.20
0.52
0.20
0.93
0.20
random evidence
full model
oracle evidence


Each part of the reader on its own (absolute values; vertical line = full model)
Retrieval: Recall@M
Retrieval: best-node IoU
Answer: contains-match
full model (reference)
ranking: no prior
ranking: no cosine
ranking: no bias terms
ranking: cosine only
ranking: prior only
graph: no hops
graph: no reasoning layers
graph: first pass only
open segment hidden
0.32
0.40
0.48
0.56
0.15
0.18
0.21
0.24
0.175
0.200
0.225
full model
one part removed or changed

Does the memory matter more for older events?

Buckets with fewer than 10 questions are left out. 867 questions; bars are 95% intervals over streams.





Contains-match accuracy by question template

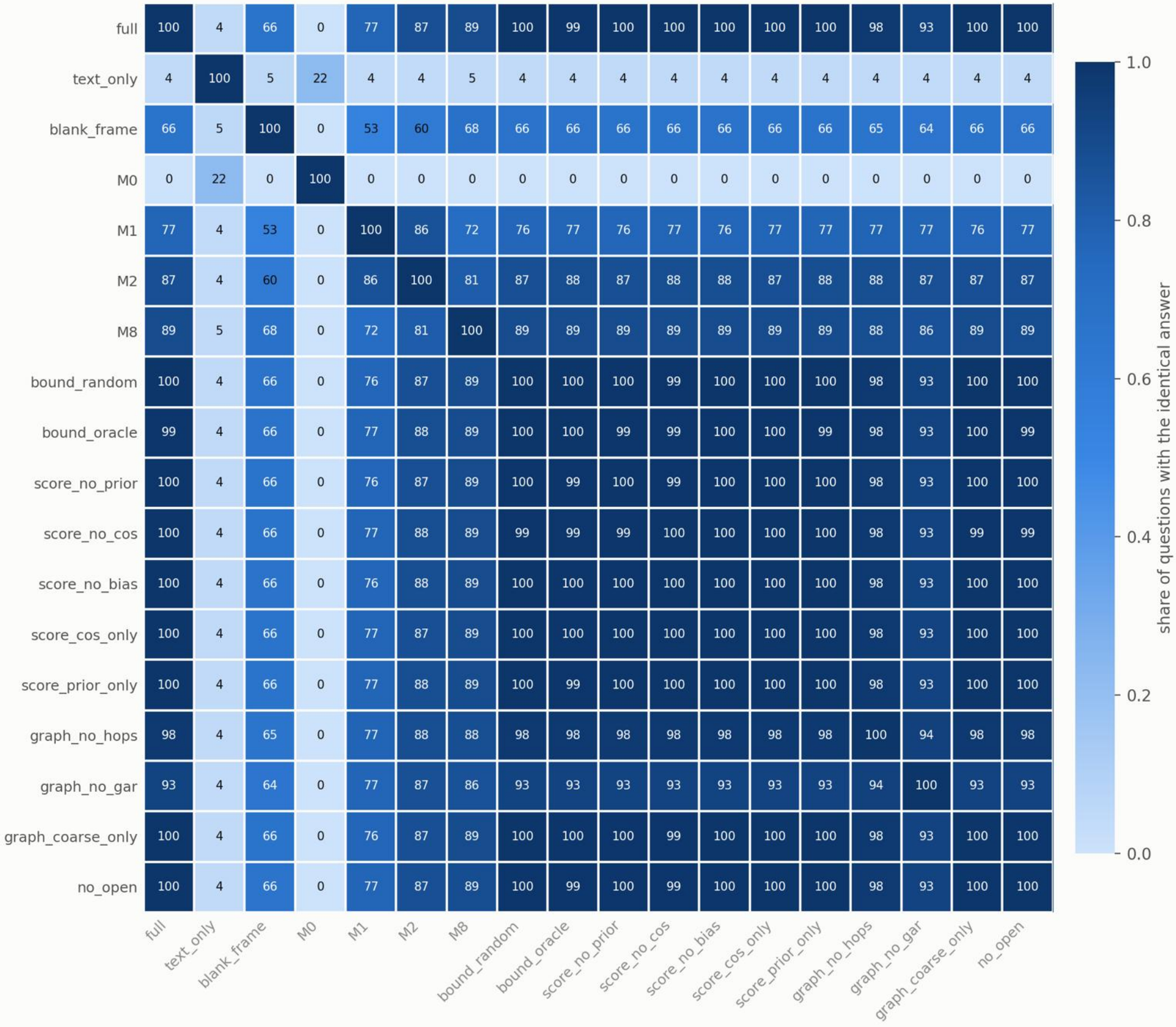
Rows are templates (question counts in brackets); columns are variants. Small templates are noisy.

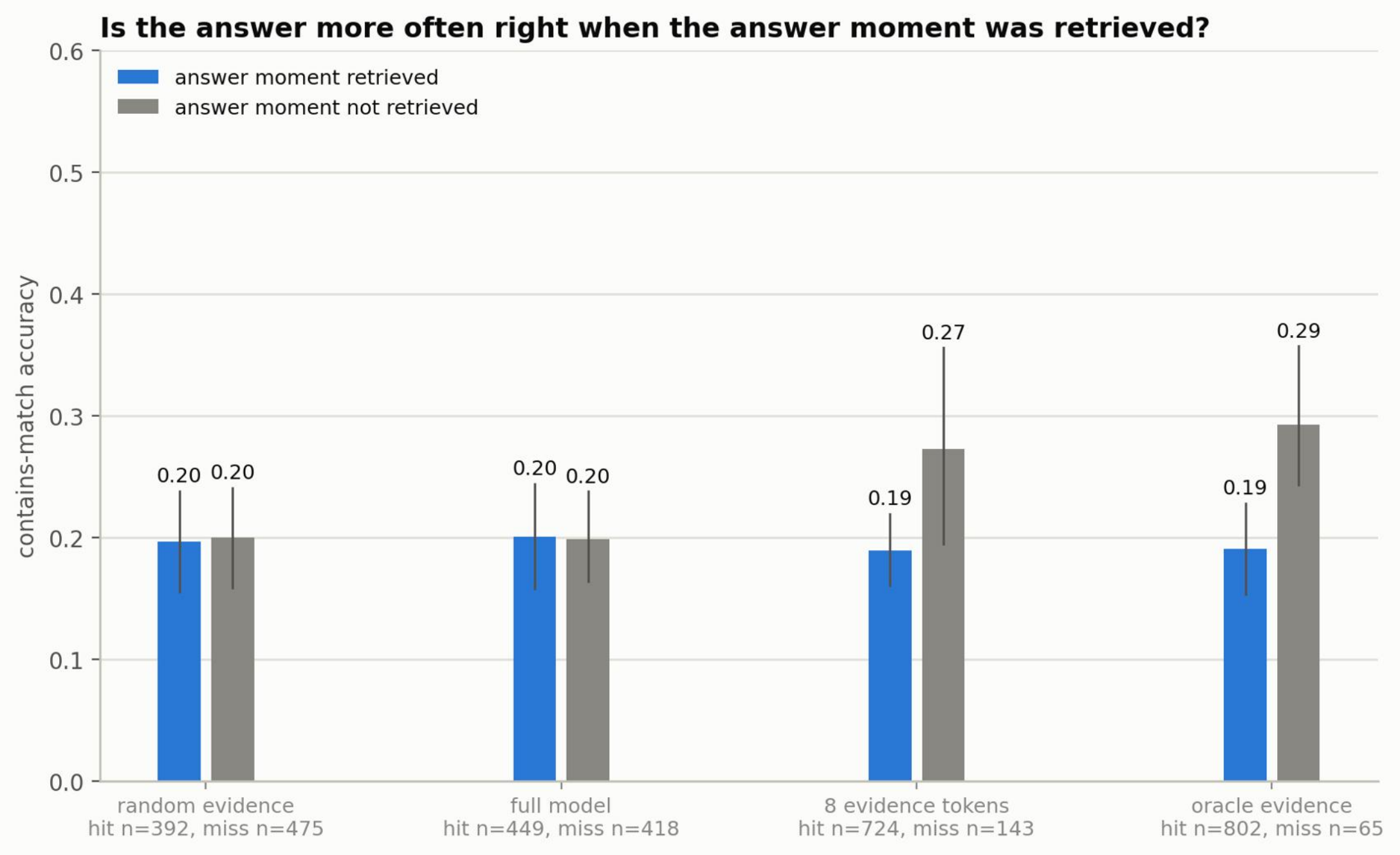
	text only	no memory	blank frame	full model	8 tokens	oracle
Objects: Where is object X before / after event Y? (n=150)	0.09	0.10	0.12	0.15	0.14	0.15
Place: Where did I put X? (n=146)	0.08	0.09	0.12	0.12	0.12	0.12
Objects: Where is object X? (n=105)	0.12	0.12	0.15	0.18	0.17	0.18
Objects: What did I put in X? (n=85)	0.08	0.11	0.14	0.14	0.14	0.14
Objects: How many X's? (quantity question) (n=73)	0.32	0.32	0.33	0.32	0.33	0.32
Objects: What X did I Y? (n=70)	0.09	0.06	0.29	0.21	0.23	0.21
Objects: In what location did I see object X ? (n=64)	0.05	0.06	0.06	0.06	0.05	0.06
Objects: What X is Y? (n=63)	0.17	0.19	0.19	0.27	0.27	0.27
Objects: State of an object (n=28)	0.36	0.46	0.57	0.57	0.61	0.57
Objects: Where is my object X? (n=20)	0.10	0.10	0.20	0.20	0.20	0.20
People: Who did I interact with when I did activity X? (n=14)	0.00	0.00	0.64	0.43	0.64	0.43
People: Who did I talk to in location X? (n=9)	0.00	0.00	0.67	0.78	0.78	0.67

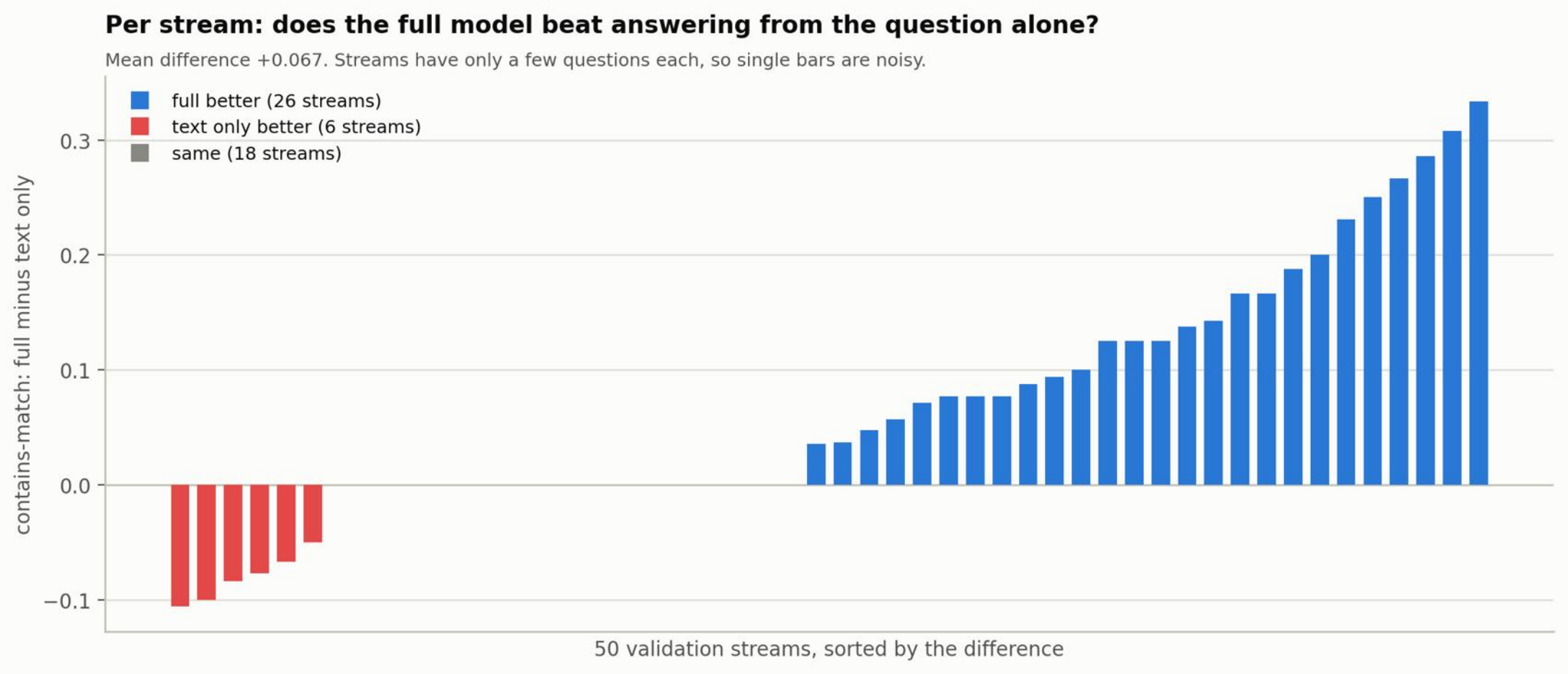


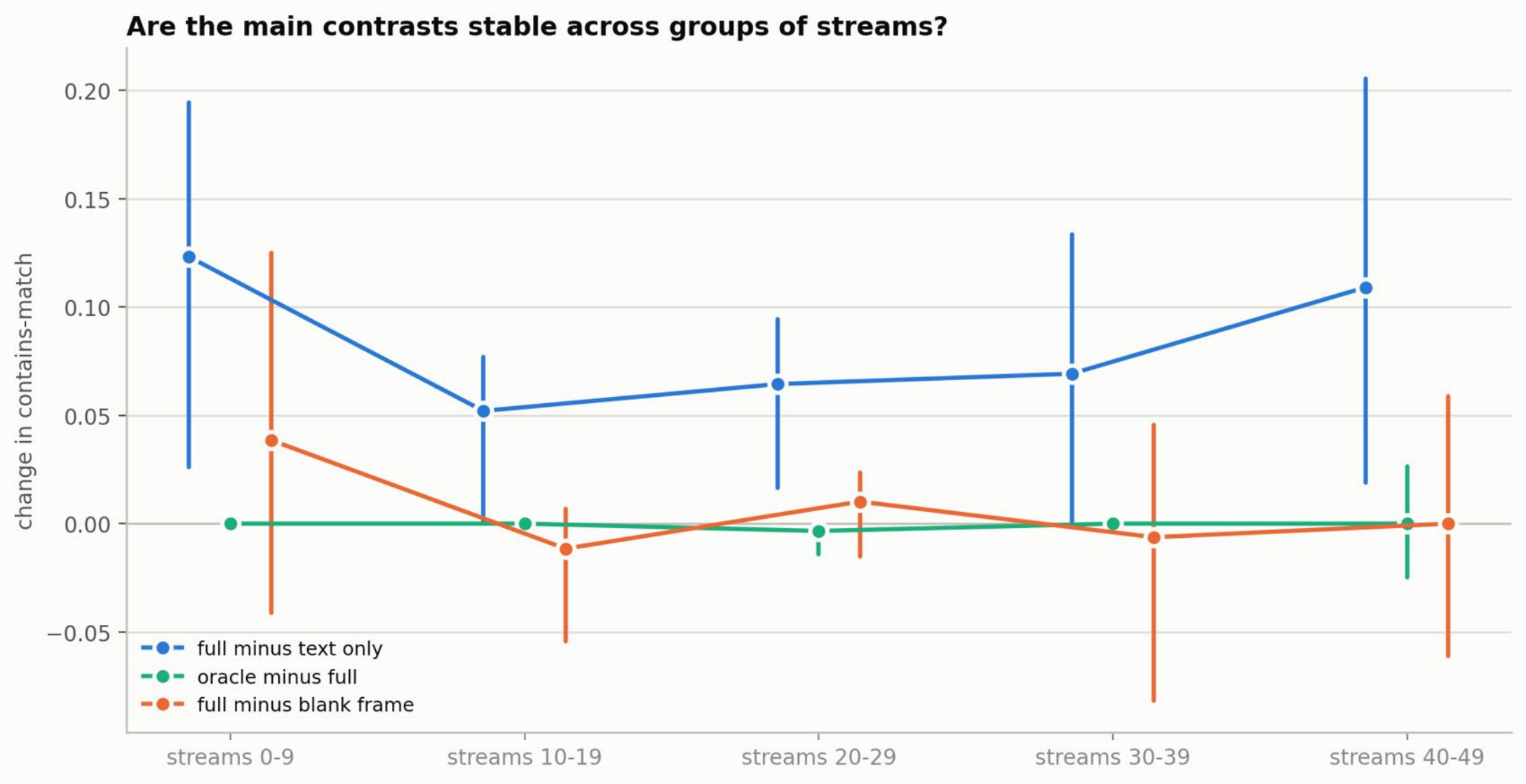
How often two variants give the same answer (percent of questions)

Against the full model, the median variant gives the identical answer on 99% of questions (lowest: M0, 0%).




Is the answer more often right when the answer moment was retrieved?
answer moment retrieved
answer moment not retrieved
contains-match accuracy
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0.20
0.20
0.20
0.20
0.19
0.27
0.19
0.29
random evidence
hit n=392, miss n=475
full model
hit n=449, miss n=418
8 evidence tokens
hit n=724, miss n=143
oracle evidence
hit n=802, miss n=65


Per stream: does the full model beat answering from the question alone?
Mean difference +0.067. Streams have only a few questions each, so single bars are noisy.
full better (26 streams)
text only better (6 streams)
same (18 streams)
contains-match: full minus text only
0.3
0.2
0.1
0.0
−0.1
50 validation streams, sorted by the difference


Are the main contrasts stable across groups of streams?
change in contains-match
0.20
0.15
0.10
0.05
0.00
−0.05
full minus text only
oracle minus full
full minus blank frame
streams 0-9
streams 10-19
streams 20-29
streams 30-39
streams 40-49

Where the answer moment sits in memory: finished (committed) vs still being written (open)

